

5⁽³⁾

Measures of Variations (Dispersions)

Coefficient of Variation

Coefficient of Variation

- ▶ One of the disadvantages of standard deviation is it depends on the unit of measurement.
- ▶ It means, it is difficult to use the standard deviation to compare measurement from different populations.

Coefficient of Variation

For this reason, statisticians have defined the *coefficient of variation*, which expresses the standard deviation as a percentage of sample or population arithmetic mean.

$$\text{i.e. } CV = \frac{\sigma}{\mu} 100\% \quad \text{or} \quad CV = \frac{s}{\bar{x}} 100\%$$

Coefficient of Variation

Notice that the numerator and the denominator in the definition of CV *have the same units*, so CV itself has no units of measurement.

This gives us the advantage of being able to compare directly the variability of two different populations using the coefficient of variation.

Coefficient of Variation

Suppose there is a population data set [80, 90, 100]. Find the mean, the population standard deviation and the coefficient of variation.

Coefficient of Variation

The mean is 90 and the population standard deviation is 8.165 and the coefficient of variation is 0.09. As a percentage, the coefficient of variation is 9%..

Coefficient of Variation

For example, a pizza restaurant measures its delivery in minutes. The average delivery time is 20 minutes and the standard deviation is 5 minutes. Find the coefficient of variation and interpret your result.

Coefficient of Variation

Example 6:- The *Tana Fishery* is a small, family-run store in Bahir Dar. This store has a limited selection of fish. In fact, the *Tana Fishery* provides eight different types of selection of fish to its customers. The price per kilo gram (in hundreds of birr) are:

2.25, 2.10, 2.15, 1.95, 2.25, 2.60, 2.00, 1.85

Compute the CV.

$$\mu = \frac{2.25+2.10+2.15+1.95+2.25+2.60+2.00+1.85}{8} = 2.14$$

Coefficient of Variation

► **Solution:-**

$$\mu = 2.14$$

$$\sigma = 0.22$$

$$CV = \frac{\sigma}{\mu} 100\% = \frac{0.22}{2.14} \times 100\%$$

$$\Rightarrow CV = 10.28\%$$

i.e. the standard deviation is 10.28% of the arithmetic mean of the given data and based on this information we can tell whether the given data are spread-out or clustered together.

Coefficient of Variation

Example 7:- The blood pressure of two patients was measured daily for three weeks. The average blood pressure of the first patient is **160** with standard deviation of **14.4** and the average blood pressure of the second patient is **136** with standard deviation of **13.6**. Which patient's blood pressure was relatively more variable? Justify.

Coefficient of Variation

Example 8:- Two plants C and D of a factory show the following results about the number of workers and the wages paid to them..

	Plant C	Plant D
Number of workers	5000	6000
Average Salary	2500	2900
Standard Deviation	9	10

Coefficient of Variation

Find the coefficient of variation and interpret your result.

In general use the coefficient of variation when you want to compare variability between

- groups that have averages of different magnitudes.
- characteristics that use different units of measurements.

Coefficient of Variation

Example 9:- Suppose you are measuring household expenditures and want to compare the variability of spending among high-income and low-income households.

Expenditure	High-income	Low-income
Mean	\$500,000	\$40,000
Standard Deviation	\$125,000	\$10,000

Chebyshev's Theorem

Chebyshev's Theorem

Chebyshev's Theorem named after the known Russian mathematician Pafnuty Chebyshev (1821-94) is a fact that applies to all possible data sets. It describes the minimum proportion of the measurements that lie within two or more standard deviations of the ***mean***.

Chebyshev's Theorem



Chebyshev's Theorem



Chebyshev's Theorem

For any population or sample, at least $1 - \frac{1}{k^2}$ of the observations in the data set fall within k standard deviations of the mean, where $k > 1$.

This theorem is valid for any type of distribution (whether *symmetric or skewed*).

Chebyshev's Theorem

This theorem says that, within **two** standard deviation of the mean (i.e. for $k = 2$) you will always find at least 75% (i.e. 75% or more) of the data.

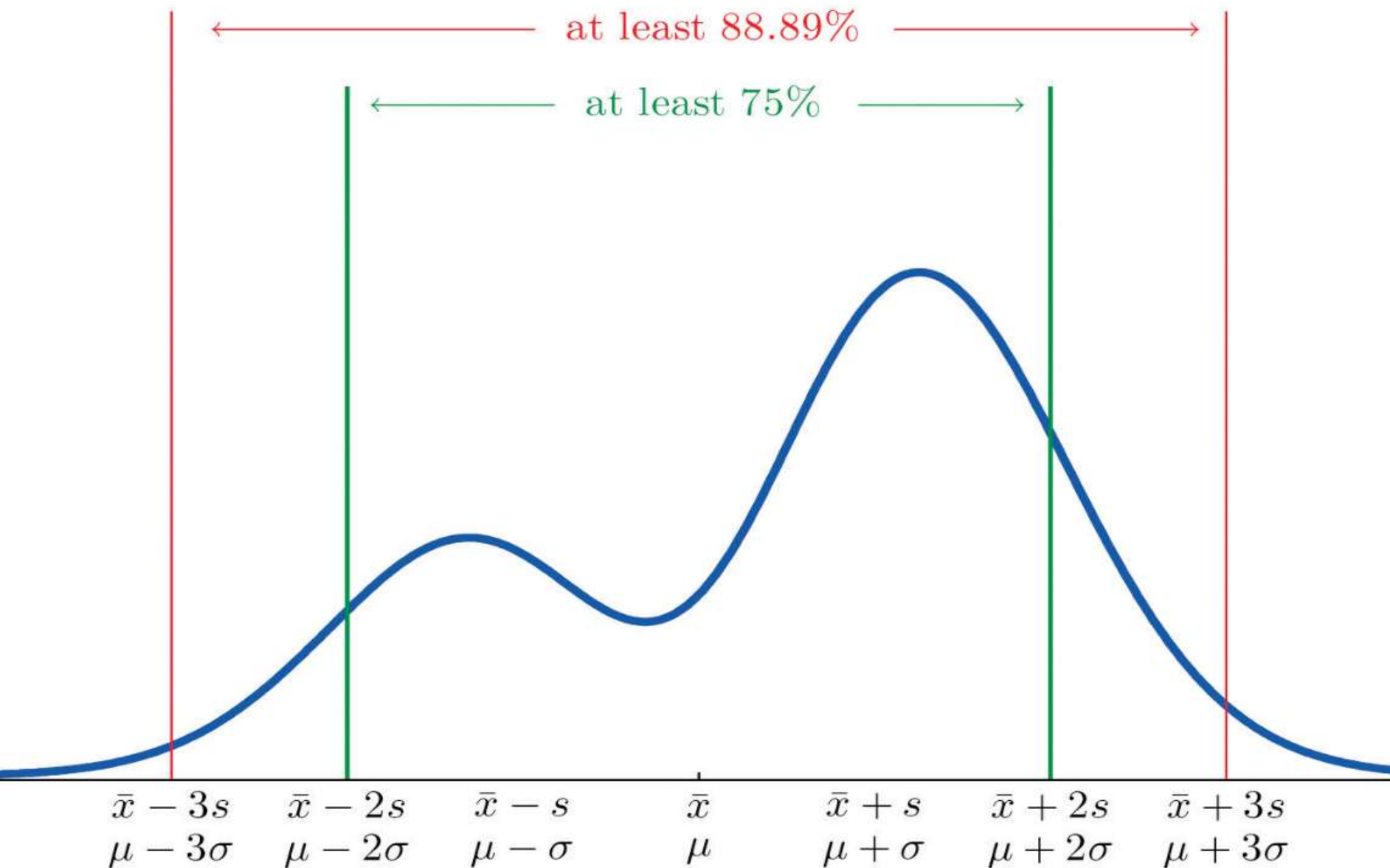
$$\begin{aligned}\text{This is because } 1 - \frac{1}{k^2} &= 1 - \frac{1}{2^2} = 1 - \frac{1}{4} = \frac{3}{4} \\ &= 0.75 = 75\%\end{aligned}$$

Chebyshev's Theorem

If we consider the interval enclosed by **three** standard deviation on either sides of the mean (i.e. $k = 3$), the theorem says that we will find that least 89% (i.e. 89% or more of the data.

$$\begin{aligned}\text{And this is because } 1 - \frac{1}{k^2} &= 1 - \frac{1}{3^2} = 1 - \frac{1}{9} = \frac{8}{9} \\ &= 0.89 = 89\%\end{aligned}$$

Chebyshev's Theorem



Chebyshev's Theorem

Example8:- Chebyshev's Theorem can be utilized for the following values of k:

i) For $k = 1.5 \Rightarrow 1 - \left(\frac{1}{1.5^2}\right) = 0.5556 = 55.56\%$

of all observations fall within 1.5 standard deviation of μ .

ii) For $k = 2.2 \Rightarrow 1 - \left(\frac{1}{2.2^2}\right) = 0.7534 = 75.34\%$ of all observations fall within 2.2 standard deviation of μ .

Chebyshev's Theorem

iii) For $k = 2.5 \Rightarrow 1 - \left(\frac{1}{2.5^2}\right) = 0.8400 = 84\%$ of all observations fall within 2.5 standard deviation of μ .

iv) For $k = 3.05 \Rightarrow 1 - \left(\frac{1}{3.05^2}\right) = 0.8925$
 $= 89.25\%$ of all observations fall within 3.05 standard deviation of μ .

Example9:- For a statistics class in spring 2025 , the average score of students was 72 and the standard deviation was 7. Using Chebyshev's theorem, find the percentage of students who scored between 58 and 86.

Any Questions



Any Questions



Thank
you